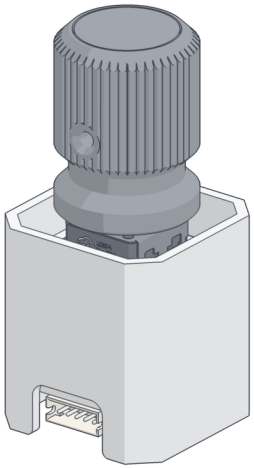
SMD BUTTONS

PP Stack

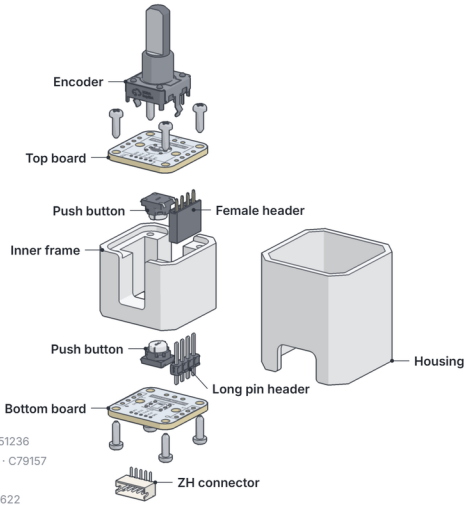
A rotary encoder you can also push and pull.



BEFORE YOU START

The parts

Two identical PCBs, a 3D-printed frame and housing, and a handful of parts to solder.



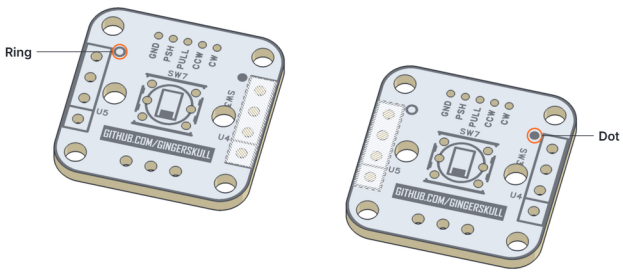
Parts · LCSC part numbers

- **2×** PCB
- **1×** Rotary encoder EC12E2420801 · C351236
- **2×** Push button Panasonic EVQ9P105M · C79157
- **1×** Female header 1×4 · C2897367
- **1×** Long pin header 1×4, 17 mm · C5156622
- **1×** ZH1.5 connector 5 pin, right angle · C2845568
- **8×** M2 × 6 self-tapping screw C357360
- **1×** Inner frame + housing printed

BEFORE YOU START

One PCB, two jobs

Both boards are the same PCB. The headers decide which is which.
U4 and U5 are mirrored, so with the top board flipped on, U4 lands on U5.



Bottom board
Long pins in U5, by the ring

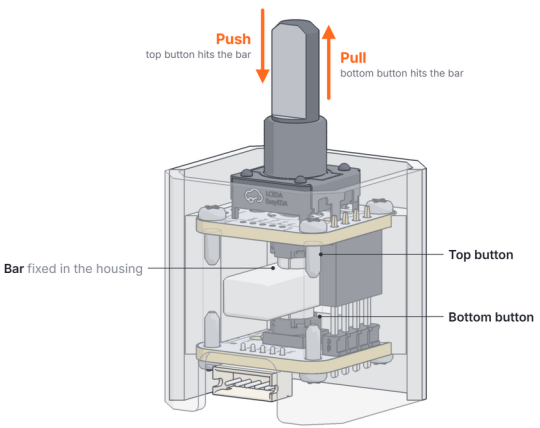
Top board
Female header in U4, by the dot

Hatched slot stays empty on each board.

BEFORE YOU START

How it works

The boards, frame and encoder move as one stack.
A bar fixed in the housing sits between the two buttons.



The buttons also spring the knob back to centre.

Parts and where to buy

github.com/gingerskull/Push-Pull-Encoder#parts-and-where-to-buy

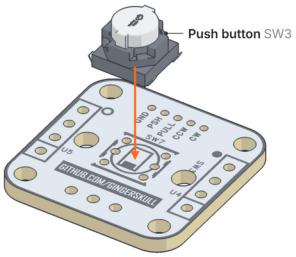
A Bottom board

Steps 1–3

1 Push button

A · Bottom board · Button side

SMD button onto its pads, centred in the SW7 outline.

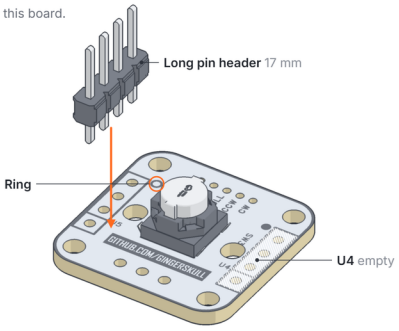


- Tack one pad first, check it sits flat, then solder the rest.
- Or reflow it on a soldering hot plate. The board is still bare.

2 Long pin header

A · Bottom board · U5

Into U5, next to the ring. U4 stays empty on this board.

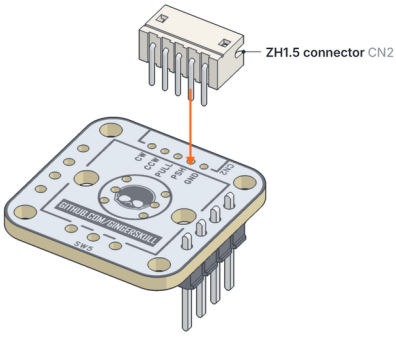


- Keep it square to the board. These pins mate with the top board later.

3 ZH connector

A · Bottom board · Flip

Flip the board. Opening faces the board edge.



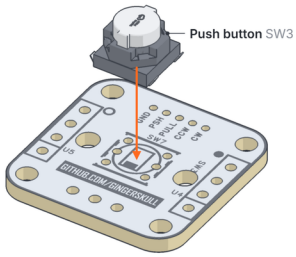
B Top board

Steps 4–6

4 Push button

B · Top board · Button side

Same as step 1: SMD button, centred in the SW7 outline.

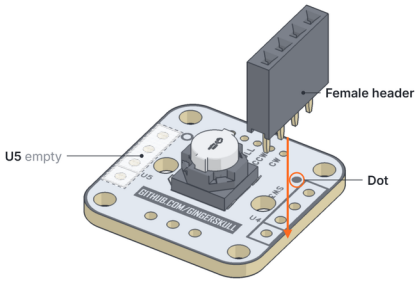


- Iron or hot plate, as in step 1. Nothing to trim.

5 Female header

B · Top board · U4

Into U4, next to the dot. U5 stays empty on this board.

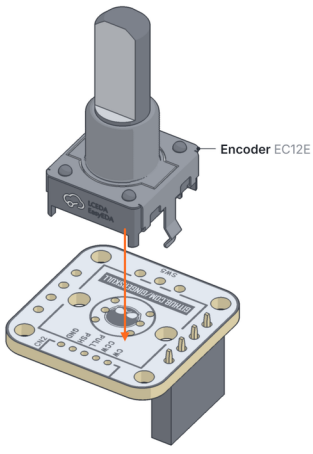


- When the boards stack, ring meets dot.

6 Encoder

B · Top board · Flip

Flip the board and fit the encoder.



- Solder the mounting tabs too. They carry the push/pull load.

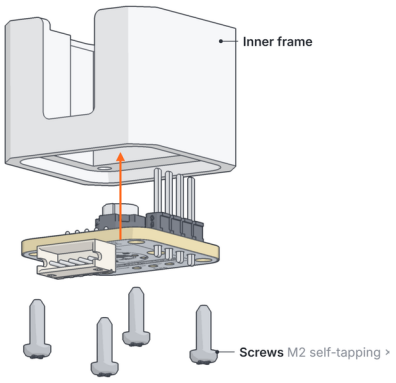
C Assembly

Steps 7–10

7 Bottom board onto the frame

C · Assembly | Bottom board + frame

Button side into the frame. Four M2 self-tapping screws.

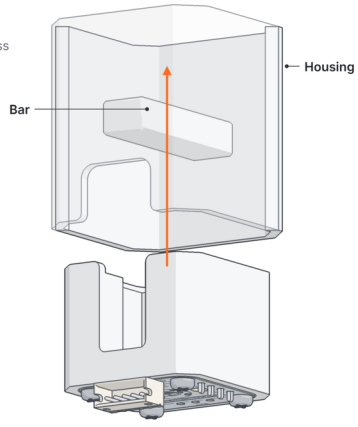


! Snug, not tight. They're cutting threads in plastic.

8 Into the housing

C · Assembly | From below

Slide the frame up into the housing. The slots pass over the bar, connector lines up with the cutout.



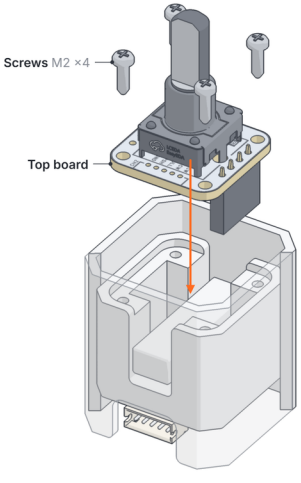
! The bar ends up between the two push buttons. That's the whole push/pull trick.

! Printed parts need only a minimal gap: smooth travel, no wobble.

9 Top board

C · Assembly | Top board, from above

Female header drops onto the long pins. Dot sits over the ring. Screw it down.

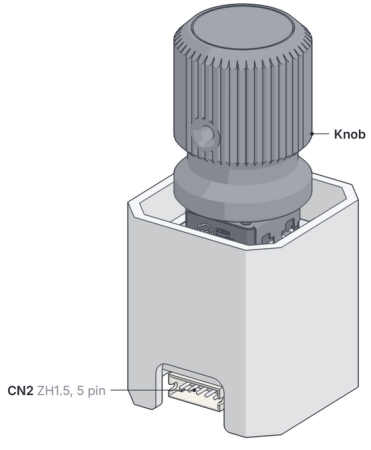


! Check the pins line up before pushing down.

10 Done

C · Assembly | PP Stack · SMD buttons

Knob on, cable in.



CN2 pinout

1. GND
2. PULL
3. PUSH
4. CCW
5. CW